Claims

1. A method for generating a physiological assessment of a user from biomechanical data gathered from the user by a system having one or more processors and at least one orthotic device, the method comprising the steps of:

acquiring biomechanical data of the user to provide mechanical and biomechanical information associated to the user with one or more biomechanical sensors of the at least one orthotic device;

generating trajectory data of the user using the acquired biomechanical data using a gait profiler and the one or more biomechanical sensors of the at least one orthotic device;

classifying the trajectory data into variables of interest data according to a set of key biomechanical features, wherein the variables of interest data are processed biomechanical data associated with specific times, postures, and/or activities;

rejecting variables of interest data that do not fit a pre-determined acceptance criteria, wherein the pre-determined acceptance criteria are cut-off requirements for accepting specific variables of interest as meaningful;

extracting key features of the non-rejected variables of interest data;

calculating a current cumulative risk data based on the extracted key features, a user profile covariate data and the variables of interest data; ~~and~~

generating the physiological assessment of the user using the current cumulative risk data; and

generating an alarm based on selected current cumulative risk data.

2. The method of claim 1, further comprising the step of:

sorting and labeling the trajectory data into discrete segments; and

filtering the discrete segments of trajectory data so as to reduce a number of individual frames and remove noise;

wherein the step of classifying the trajectory data into variables of interest data according to a set of key biomechanical features is performed on the filtered discrete segments of trajectory data.

3. The method of either one of claims 1 or 2, further comprising the step of formatting and simplifying the risk data of the assessment of the user for presentation to the user.

4. The method of claim 3, further comprising the step of adding biomechanically derived information to the assessment of the user.

5. The method of any one of claims 1 to 4, wherein the key biomechanical features are obtained by forming an indexed combination of curated data from a reference database and an outcome database:

the reference database containing key biomechanical measurements and associated physiological determinant factors, and being constructed using peer-reviewed academic publications; and

the outcome database containing linked key biomechanical measurements and associated statistically established outcomes, and being constructed using user monitoring and experimentation.

6. The method of any one of claims 1 to 5, wherein the biomechanical data of the user is acquired from the biomechanical sensors positioned on the user.

7. The method of claim 6, wherein the biomechanical sensors include inertial and angular sensors positioned on a lower-body orthotic device worn by the user.

8. A system for generating a physiological assessment of a user from biomechanical data gathered from the user, comprising:

biomechanical sensors configured to observe associated user body segment kinematics;

a processor in communication with the plurality of biomechanical sensors of at least one orthotic device, the processor having an associated memory comprising instructions stored therein that when executed on the processor perform the steps of:

receiving biomechanical data of the user from the plurality of biomechanical sensors of the at least one orthotic device to provide mechanical and biomechanical information associated to the user;

generating trajectory data of the user using the acquired biomechanical data using a gait profiler and the plurality of biomechanical sensors of the at least one orthotic device;

classifying the trajectory data into variables of interest data according to a set of key biomechanical feature, wherein the variables of interest data are processed biomechanical data associated with specific times, postures, and/or activities s;

rejecting variables of interest data that do not fit a pre-determined acceptance criteria, wherein the pre-determined acceptance criteria are cut-off requirements for accepting specific variables of interest as meaningful;

extracting key features of the non-rejected variables of interest data;

calculating a current cumulative risk data based on the extracted key features, a user profile covariate data and the variables of interest data; ~~and~~

generating the physiological assessment of the user using the current cumulative risk data; and

generating an alarm based on selected current cumulative risk data.

9. The system of claim 8, wherein the memory comprises further instructions stored therein that when executed on the processor further perform the steps of:

sorting and labeling the trajectory data into discrete segments; and

filtering the discrete segments of trajectory data so as to reduce a number of individual frames and remove noise;

wherein the step of classifying the trajectory data into variables of interest data according to a set of key biomechanical features is performed on the filtered discrete segments of trajectory data.

10. The system of either one of claims 8 or 9, wherein the memory comprises further instructions stored therein that when executed on the processor further perform the steps of formatting and simplifying the risk data of the assessment of the user for presentation to the user.

11. The system of claim 10, wherein the memory comprises further instructions stored therein that when executed on the processor further perform the steps of adding biomechanically derived information to the assessment of the user.

12. The system of any one of claims 8 to 11, further comprising:

a reference database containing key biomechanical measurements and associated physiological determinant factors, from peer-reviewed academic publications; and

an outcome database containing linked key biomechanical measurements and associated statistically established outcomes, from user monitoring and experimentation;

wherein the key biomechanical features are obtained by forming an indexed combination of curated data from the reference database and the outcome database.

13. The system of any one of claims 8 to 12, wherein the biomechanical sensors include inertial and angular sensors positioned on a lower-body orthotic device worn by the user.

14. The system of any one of claims 8 to 12, further comprising a lower-body orthotic device configured to be worn by the user, wherein the biomechanical sensors include inertial and angular sensors positioned on the lower-body orthotic device.